A Study on Enhancing Hydrogen Storage in Calcium Hydride through Ball Milling

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Abstract: This research aims to explore the impact of ball milling on calcium hydride's effectiveness as a hydrogen storage medium. With the pressing need to transition from fossil fuels to cleaner energy sources, hydrogen presents a viable alternative due to its high energy yield and environmentally benign combustion products. Metal hydrides, particularly calcium hydride, offer a promising avenue for hydrogen storage. However, the efficiency of hydrogen desorption and absorption in calcium hydride can be further enhanced. This study seeks to investigate how ball milling, a process known to modify physical properties, affects the hydrogen storage capabilities of calcium hydride.

Keywords: Ball milling, Hydrogen Storage, Calcium Hydride

1 Introduction

The transition from fossil fuels to renewable energy sources is imperative to mitigate environmental degradation and climate change[1, 2, 3]. Hydrogen, with its high energy content and clean combustion, stands out as a leading alternative. Among the various hydrogen storage methods, metal hydrides offer a compact and safe solution[4, 5]. This research focuses on calcium hydride, a material known for its high hydrogen storage capacity but limited by kinetic and thermodynamic barriers. Previous studies have extensively explored composite hydrides but rarely delved into pure calcium hydride. We hypothesize that ball milling could significantly improve the hydrogen storage performance of calcium hydride by altering its surface properties and particle size, potentially making it a more efficient storage medium.

2 Related Works

Various studies have highlighted the potential of ball milling to enhance hydrogen storage in metal hydrides. For instance, research on $Mg_{17}Al_{12}$ hydride with CaH_2 showed significant improvements in hydrolysis reactivity, achieving a hydrogen generation yield of 94.8% under optimal conditions[6]. Another study on $MgH_2 - Ca$ and $MgH_2 - CaH_2$ mixtures revealed that ball-milling leads to faster hydrolysis reactions and higher hydrogen yields, with a notable reaction yield of 80% after 30 minutes of hydrolysis[7]. Additionally, the investigation of the $CaH_2 + MgB_2$ system with AlB_2 additions demonstrated improved kinetics and hydrogen sorption properties, suggesting the importance of the milling process and additive selection in enhancing hydrogen storage capabilities[8]. Lastly, the synthesis of calcium amidoborane through reactive milling exhibited complex hydrogen desorption behavior, underscoring the nuanced effects of structural modifications on storage performance[9]. These findings collectively suggest that ball milling is a promising technique for improving the hydrogen storage characteristics of metal hydrides, including pure calcium hydride, by modifying surface properties and particle sizes. However, these studies predominantly focus on composite materials, leaving a gap in the research regarding the effects on pure calcium hydride. This proposal aims to fill that gap.

3 Methodology

The proposed research project is designed to conduct a comprehensive and systematic investigation into the effects of mechanical milling on the hydrogen storage properties of calcium hydride (CaH_2). This study aims to explore the fundamental mechanisms by which ball milling, a process of mechanical alloying, influences the microstructural, morphological, and hydrogen storage characteristics of CaH_2.

The research methodology will involve a series of controlled experimental procedures, where calcium hydride samples will be subjected to ball milling under various conditions. The variables of interest include milling time (ranging from a few hours to several days to assess the impact of prolonged mechanical energy input), ball-to-powder ratio (to determine the optimal mass ratio for efficient milling), and the atmosphere in which milling is conducted (to investigate the effects of environmental conditions such as air, inert gas, or vacuum on the stability and properties of milled CaH_2).

To ensure a comprehensive analysis of the milled samples, an array of analytical techniques will be employed. X-ray diffraction (XRD) will be utilized to identify

any phase transformations or crystallite size reductions resulting from the mechanical milling process. Scanning electron microscopy (SEM) will provide detailed images of the surface morphology and particle size distribution, offering insights into the physical changes induced by milling. Hydrogen sorption measurements, including both absorption and desorption kinetics, will be conducted to evaluate the practical hydrogen storage capacities and the efficiency of hydrogen uptake and release. These measurements will be crucial in assessing the potential of milled CaH_2 for practical hydrogen storage applications.

Furthermore, additional analytical methods such as transmission electron microscopy (TEM) for more detailed structural analysis, BET surface area measurements for assessing changes in surface area, and thermogravimetric analysis (TGA) under hydrogen atmosphere for thermal stability assessments may also be incorporated into the research design.

4 Conclusion

We anticipate that ball milling will enhance the hydrogen storage capabilities of calcium hydride by creating more reactive surfaces and reducing particle size, thereby improving the kinetics of hydrogen absorption and desorption. This research could pave the way for the development of more efficient hydrogen storage materials, contributing to the broader adoption of hydrogen as a clean energy carrier.

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